DS4420 Final Project

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Topic: MLP and Bayesian Models
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Beyond the Budget: Predicting Box Office Performance Using Production Features and Digital Signals

Purpose

This project investigates whether pre-release digital signals improve predictions of movie box office revenue beyond what traditional production-level features can achieve.

We compare two complementary modeling approaches:

1. A Bayesian linear regression model implemented by hand in R. Uses Metropolis-Hastings MCMC for posterior distributions.
2. A Multilayer Perceptron implemented in Python, using NumPy for the model, and sklearn for preprocessing, to capture nonlinear relationships.

Each model is trained on traditional features alone (such as runtime, budget, genre, actors, popularity, etc.), as well as an enriched dataset which adds Google Trends data.

Background & Data

Background:

The global film industry generated over $33 billion in box office revenue in 2025 [1]. The industry suffered significantly during the Covid-19 pandemic, and the recent Writers Guild of America strike. As revenue has been steadily increasing towards Pre-pandemic levels, predicting box office performance is still notoriously difficult [5]. The intent of our project was to look at a combination of production features and social discourse around movies, to see if a combined approach would enhance movie box office predictions [4]. Since the Twitter API is no longer available for free use, we were limited to just using Google Trends for our social discourse measure. We continued the project using just Google Trends, though it is a less comprehensive signal than we had initially planned.

Dataset:

TMBD 5000 Movie Dataset from Kaggle, filtered to 2276 movies (post 2000, valid budget, revenue, and runtime)

Traditional (Production) Features:

Log budget, runtime, popularity, vote average, vote count, number of genres, number of cast, major studio indicator, number of production companies, summer release, holiday release, franchise, English indicator, 12 genres one-hot encoded

Enriched Feature:

Google Trends, pre-release search interest

Target:

Log-transformed box office revenue

Preprocessing:

Min-Max Scaled (MLP) or z-score standardized (Bayesian) with bias term, 80/20 train test split.

Methods

Multilayer Perceptron (MLP)

Implemented in Python with NumPy for the model and sklearn for preprocessing (MinMaxScaler, train/test split).

Architecture:
Input → 32 unit hidden layer with ReLU activation → Output

Training:
Full-batch gradient descent, 500 epochs, n = 0.001

Bayesian Linear Regression

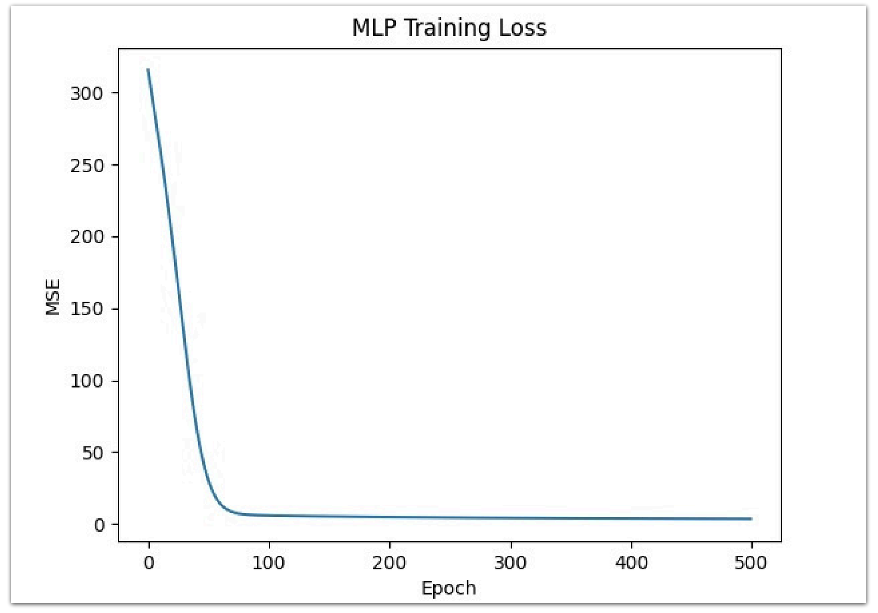
Fully implemented by hand in R with no external packages. Uses Metropolis-Hastings MCMC for posterior interface.

Priors:
$\beta \sim N(0,$
$\sigma^2 \sim$ Inverse-Gamma(a_0=2, b_0=1)

Sampling:
10,000 MCMC iterations, 2,000 burn in. Log-space proposals for σ^2 with Jacobian correction. Thinned every 10th sample. Convergence assessed via trace plots and ACF. Posterior predictive intervals for uncertainty quantification/.

Results

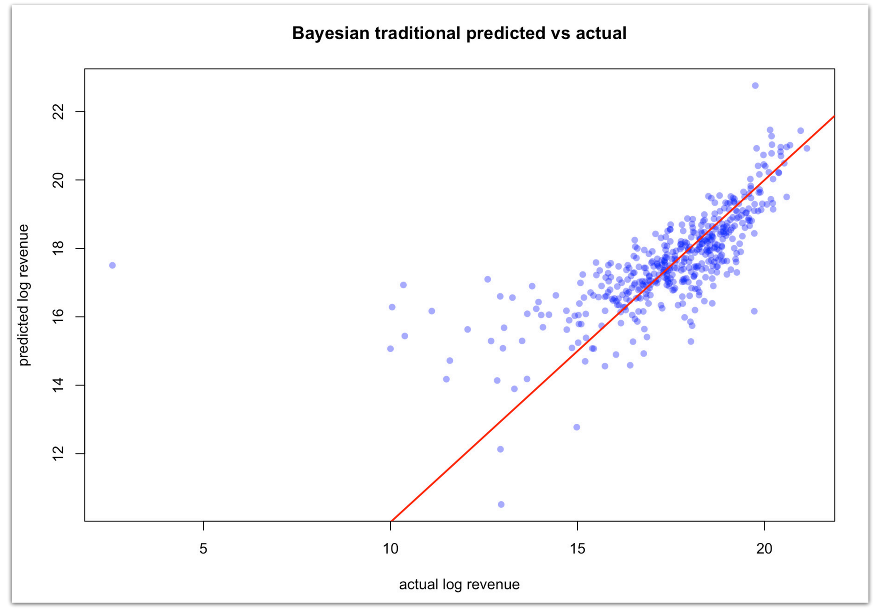
MLP Results



	Traditional	Enriched
RMSE	1.8116	1.8803
MAE	1.2391	1.3151
R^2	0.2240	0.1640

The Google Trends did not improve the MLP model. R^2 dropped from 0.224 to 0.164, suggesting that the signal adds noise at this dataset scale.

Bayesian Results



	Traditional	Enriched
RMSE	1.3517	1.3493
MAE	0.8196	0.8179
R^2	0.5246	0.5263

The Google Trends had a marginal improvement on the Bayesian model. The R^2 value increased from 0.5246 to 0.5263. The strongest predictors were log_budget, popularity, and vote_count.

Conclusions

The Bayesian model (R^2=0.525) substantially outperformed the MLP (R^2=0.224), suggesting that for this dataset, a well defined linear model with uncertainty quantification beats a shallow neural network.

Budget, popularity, and vote count emerged as the strongest predictors across both approaches.

Google Trends had opposite effects: a marginal improvement for the Bayesian model (R^2 0.5246 → 0.5263) but a decline for MLP (R^2 0.224 → 0.164), indicating the linear model better absorbs a week signal while the MLP over-fits to its noise.

The Bayesian model's results indicate that digital signals have a slight positive impact on predictive accuracy, warranting exploration of stronger digital signals. The Bayesian approach's posterior credible intervals also offer practical value for studios quantifying forecast uncertainty.

Future Work

The results of our Bayesian model indicated that the digital signal had slight improvement in predictive accuracy. A future work consideration to expand on our findings would be to explore other, potentially stronger, digital signals. The Twitter API would provide valuable insight to social discourse trends and complement the Google Trends data.

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